ESEMPIO: Codifica TRUECOLOR, RGB

24 BIT : 8 bit, 8 bit, 8 bit $\rightsquigarrow N_b = 24$

ROSSO, VERDE, BLU

$0 \div 255$, $0 \div 255$, $0 \div 255$

$2^{24} \approx 16$ Milioni di colori

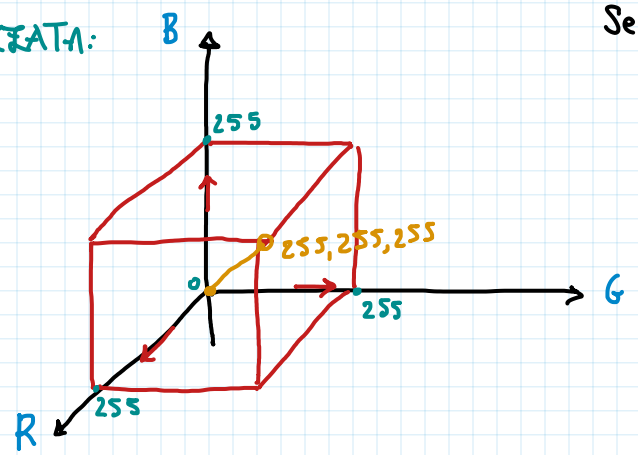
Se proietto sulla diagonale:

$\tilde{f} : \mathbb{R}^2 \longrightarrow \mathbb{R}$

SCALA DI GRIGI

IMMAGINE DIGITALIZZATA:

$$I \begin{cases} R \in \mathbb{R}^{H \times W} \\ G \in \mathbb{R}^{H \times W} \\ B \in \mathbb{R}^{H \times W} \end{cases}$$



Def: Si dice TENSORE una applicazione MULTI-LINEARE a valori in $\mathbb{R}$.

$$T : \underbrace{V \times V \times V \times \ldots \times V}_{k\text{-volte}} \times \underbrace{V^* \times \ldots \times V^*}_{h \text{ volte}} \longrightarrow \mathbb{R}$$

dove V è uno spazio vettoriale, e V^* suo duale

$$V^* = \{ \underset{\text{lineari}}{f : V \to \mathbb{R}} \}$$

ESEMPIO:

Una matrice $M \in \mathbb{R}^{H \times W}$ si def. $M : V \times V \longrightarrow \mathbb{R}$

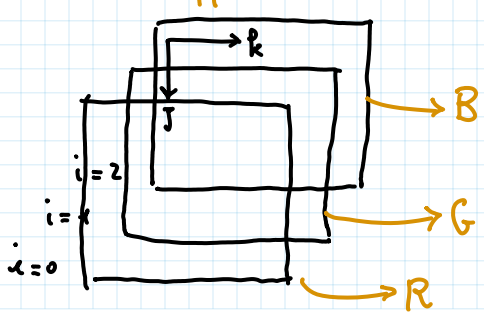
$$(i, j) \longmapsto m_{ij} \in \mathbb{R}$$

Quindi $M \in V^* \times V^* = \{ \underset{\text{lineari}}{f : V \times V \to \mathbb{R}} \}$

ESEMPIO: $T : V \times (V^* \times V^*) \longrightarrow \mathbb{R}$

$$(i, \underbrace{j, k}_{M}) \longmapsto T^i_{jk}$$

$T =$



CHANNEL / CANALE

$$T^{1}_{34} = G_{34}$$

coordinate del PIXEL